

Digital Evidence & Forensic Incident Dossier

REPORT REFERENCE:	RPT-█20260907-█DC2F3E	GENERATED (UTC):	2026-█09-█07 20:█20:█22 UTC
EVIDENCE IDENTIFIER:	sample-█001	ASSIGNED CASE:	UNASSIGNED / █ DIRECT EVIDENCE
INVESTIGATOR:	SOC Lead Analyst	CLASSIFICATION:	CONFIDENTIAL // EVIDENTIARY DISCLOSURE
EMAIL SHA-256:	97d4b2e811c7520e5e79603f9050█d268159b360b9432df03d4083d8e█57ef228a	HASH ALGORITHM:	FIPS PUB 180-4 SHA-256 (Preserved Literal)

1. Executive Forensic Summary

DETERMINISTIC THREAT SCORE	ASSESSED SEVERITY	ML PHISHING PROBABILITY	ML CLASSIFICATION
82/100	CRITICAL	89.0%	PHISHING

Email risk assessed as Critical (82/100) driven by 5 forensic signal(s): Brand impersonation (+18), Credential harvesting (+12), DMARC failure (+12), Link mismatch (+15), NLP phishing classifier (+10).

2. Email Headers & Sender Alignment

From:	Microsoft Security &█security@█microsoft-█example.█com&█;
To:	employee@█company.█com
Subject:	URGENT:█ Verify your Microsoft Account
Date:	Sun,█ 7 Sep 2026 10:█42:█00 +█0530
Reply-To:	support@█example.█net
Return-Path:	bounce@█example.█net
Message-ID:	&█;█abc123@█example.█com&█;

From Domain:	microsoft-█example.█com	Reply-To Domain:	example.█net (MISMATCH - SUSPICIOUS)
Return-Path Domain:	example.█net (MISMATCH)	Alignment Verdict:	REPLY-TO / SENDER SPOOFING DETECTED

3. Email Authentication Findings (SPF / DKIM / DMARC)

PROTOCOL	OBSERVED VERDICT	TECHNICAL DETAILS & EVIDENCE
SPF (Sender Policy)	FAIL	spf=█fail (domain of bounce@█example.█net does not designate 203.█0.█113.█25)
DKIM (Cryptographic)	FAIL	dkim=█fail header.█i=█@█company.█com
DMARC (Domain Policy)	FAIL	dmARC=█fail (p=█reject dis=█none)

Notice: Observed authentication result from supplied headers (unverified by local mail server)

4. Deterministic Threat Score Breakdown

CATEGORY	IMPACT	FORENSIC FINDING & JUSTIFICATION
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General	+18 pts	N/A
General	+15 pts	N/A
General	+12 pts	N/A
General	+12 pts	N/A
General	+10 pts	N/A
General	+10 pts	N/A
General	+8 pts	N/A
General	+8 pts	N/A

5. NLP Content Phishing Classification

Model Name:	TF-IDF + Logistic Regression	Classification:	PHISHING
Phishing Probability:	89.0%	Model Confidence:	high
Top Text Features:	httpurl, verify, password, update, account	Signal Role:	Auxiliary ML signal (Non-overriding)

6. AI Analyst Assistant Synthesis

Likely Attack Type:	Credential Harvesting / Brand Impersonation Spear-Phishing	Likely Objective:	Exfiltrate corporate Microsoft credentials via disguised credential-collection landing pages.
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Executive Synthesis: Suspicious message exhibiting critical credential-harvesting indicators with severe sender spoofing, including failed DMARC and SPF checks alongside a deceptive lookalike domain.

Key Forensic Evidence Identified by LLM:

- Failed DMARC (p=reject) and SPF authentication from sending IP 203.0.113.25
- Lookalike domain micro\$oft-example.com imitating Microsoft (registered 18 days ago)
- HTML display anchor showing microsoft.com while pointing to evil-tracker.example
- NLP Phishing Classifier flagged high confidence (89% probability) for credential harvesting language
- Suspicious embedded shortened link (bit.ly/3sample-invoice)

7. Extracted Indicators of Compromise (IOCs)

Observed URLs & Suspicion Scoring:

EXTRACTED URL / TARGET HOST	SCORE	FORENSIC OBSERVATIONS & FLAGS
https://micro\$oft-example.com/login	25/100	Potential brand impersonation: similar to microsoft.com (91% match). Suspicious security/authentication keywords found: login
http://evil-tracker.example/login	40/100	HTML display link mismatch: visible text claims 'microsoft.com' but destination links to 'evil-tracker.example'. Suspicious security/authentication keywords found: login
https://bit.ly/3sample-invoice	20/100	URL shortener domain detected (bit.ly). Suspicious security/authentication keywords found: invoice

Observed Attachments & Cryptographic Checksums:

FILENAME	MIME TYPE	SIZE	SHA-256 CHECKSUM
invoice.█pdf	application/█pdf	235.4 KB	e3b0c44298fc1c149afb4c8█996fb92427ae41e4649b934c█a495991b7852b855

8. Network Transmission Path & Topology

Earliest Observable Sending IP:	198.█51.█100.█12	Reverse DNS / Host:	smtp.█example.█org
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Observed Infrastructure Route (Chronological Transmission Sequence):

HOP	FROM (HOST / IP)	BY (RECEIVING RELAY)	TIMESTAMP
#1	smtp.█example.█org (198.█51.█100.█12)	mail.█example.█net (no IP)	Sun,█ 7 Sep 2026 10:█41:█59 +█0530
#2	mail.█example.█net (203.█0.█113.█25)	mx.█company.█com (no IP)	Sun,█ 7 Sep 2026 10:█42:█01 +█0530

Notice: Labeled strictly as observed infrastructure route. Does not represent physical attacker travel path.

9. Infrastructure & Domain Intelligence

IP ADDRESS	LOCATION (CITY/COUNTRY)	ASN / NETWORK	ORGANIZATION / ISP
203.█0.█113.█25	Amsterdam,█ Netherlands	AS12345	Example Cloud Services
198.█51.█100.█12	Amsterdam,█ Netherlands	AS12345	Example Cloud Services

10. Brand Impersonation & Lookalike Domains

SUSPECTED BRAND	SPOOFED DOMAIN	SIMILARITY	TECHNIQUES & ATTACK PROFILE
Microsoft	micros0ft-█example.█com	91.0%	character_█substitution,█ brand_█keyword

11. Prescriptive Recommendations & Containment Actions

- [] 1. Block sending IP 203.0.113.25 and domain micros0ft-example.com on perimeter email gateway
- [] 2. Revoke active sessions and force password resets for any recipient who accessed the link
- [] 3. Search SIEM proxy/DNS telemetry for outbound HTTP requests to evil-tracker.example and bit.ly/3sample-invoice
- [] 4. Purge matching message IDs from all internal Exchange/M365 mailboxes

12. Evidentiary Limitations & Chain-of-Custody Certification

Forensic Chain-of-Custody Certification:
This document is generated by MailTrace AI based on literal RFC-822 evidence headers. Public IP geolocation represents coarse regional estimates based on RIR allocations and does not establish street-level physical locations or the personal identity of the sender. All cryptographic hash sums (SHA-256) were computed over unaltered message byte streams and preserve complete digital non-repudiation chain of custody.

EVIDENTIARY DIGITAL SEAL:	SHA256: 97d4b2e811c7520e5e79603f9050 █d268159b360b9432df03d4083d8e█57ef228a	VERIFICATION STATUS:	CRYPTOGRAPHICALLY VERIFIED
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